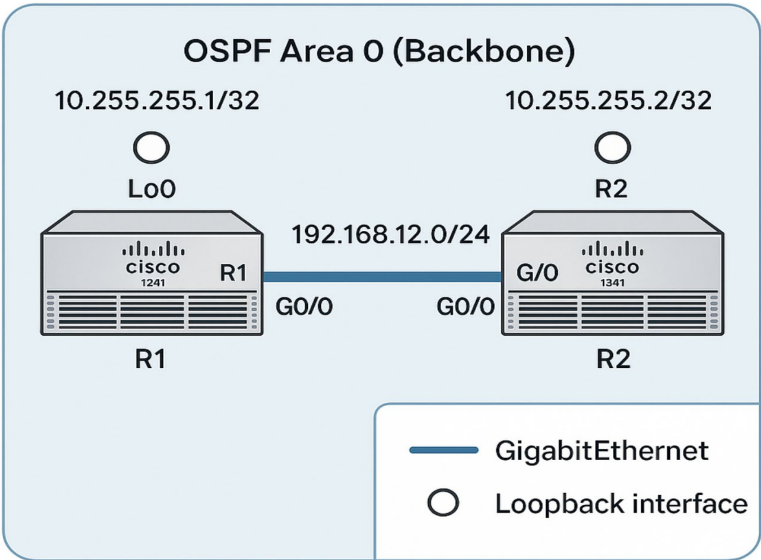


OSPF Basic Configuration Lab

Lab Objective

This lab exercise will teach students how to configure and verify Open Shortest Path First (OSPF) routing protocol between two Cisco 1941 routers. Upon completion, students will understand basic OSPF configuration commands, network advertisement, and verification procedures.

Topology Overview



- 2 Cisco 1941 routers (R1 and R2)
- Direct connection between routers using Gigabit Ethernet interfaces
- Each router will have a loopback interface for OSPF router ID stability
- All networks will be in OSPF Area 0 (backbone area)

IP Addressing Table

Device	Device Name	Device Name	Device Name	Device Name
R1	GigabitEthernet0/0	192.168.12.1	255.255.255.0	Connection to R2
R1	Loopback0	10.255.255.1	255.255.255.255	OSPF Router ID
R1	Internal Network	192.168.1.1	255.255.255.0	Internal network interface
R2	GigabitEthernet0/0	192.168.12.2	255.255.255.0	Connection to R1
R2	Loopback0	10.255.255.2	255.255.255.255	OSPF Router ID
R2	Internal Network	192.168.2.1	255.255.255.0	Internal network interface

Step-by-Step Configuration

Step 1: Basic Router Configuration

First, configure hostnames and IP addresses on both routers:

Router 1:

```
Router>enable
Router#configure terminal
Router(config)#hostname R1
R1(config)#interface GigabitEthernet0/0
R1(config-if)#ip address 192.168.12.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#interface loopback 0
R1(config-if)#ip address 10.255.255.1 255.255.255.255
R1(config-if)#exit
```

Router 2:

```
Router>enable
Router#configure terminal
Router(config)#hostname R2
R2(config)#interface GigabitEthernet0/0
R2(config-if)#ip address 192.168.12.2 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#interface loopback 0
R2(config-if)#ip address 10.255.255.2 255.255.255.255
R2(config-if)#exit
```

Step 2: Enable OSPF Process

Enable OSPF routing process on both routers using process ID 1:

Router 1:

```
R1(config)#router ospf 1
R1(config-router)#router-id 10.255.255.1
```

Router 2:

```
R2(config)#router ospf 1
R2(config-router)#router-id 10.255.255.2
```

Step 3: Configure Network Statements

Add network statements to advertise connected networks:

Router 1:

```
R1(config-router)#network 192.168.12.0 0.0.0.255 area 0
R1(config-router)#network 10.255.255.1 0.0.0.0 area 0
R1(config-router)#network 192.168.1.0 0.0.0.255 area 0
```

Router 2:

```
R2(config-router)#network 192.168.12.0 0.0.0.255 area 0
R2(config-router)#network 10.255.255.2 0.0.0.0 area 0
R2(config-router)#network 192.168.2.0 0.0.0.255 area 0
```

Step 4: Verify OSPF Configuration

Use the following commands to verify OSPF operation:

```
show ip ospf neighbor
show ip ospf interface
show ip route
show ip ospf database
```

Expected output for show ip ospf neighbor should show the neighboring router:

```
R1#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.255.255.2	1	FULL/DR	00:00:36	192.168.12.2	GigabitEthernet0/0

Complete Router Configurations

Router 1 (R1) Complete Configuration:

```
hostname R1
!
interface GigabitEthernet0/0
 ip address 192.168.12.1 255.255.255.0
 no shutdown
!
interface Loopback0
 ip address 10.255.255.1 255.255.255.255
!
router ospf 1
 router-id 10.255.255.1
 network 192.168.12.0 0.0.0.255 area 0
 network 10.255.255.1 0.0.0.0 area 0
 network 192.168.1.0 0.0.0.255 area 0
!
end
```

Router 2 (R2) Complete Configuration:

```
hostname R2
!
interface GigabitEthernet0/0
 ip address 192.168.12.2 255.255.255.0
 no shutdown
!
interface Loopback0
 ip address 10.255.255.2 255.255.255.255
!
router ospf 1
 router-id 10.255.255.2
 network 192.168.12.0 0.0.0.255 area 0
 network 10.255.255.2 0.0.0.0 area 0
 network 192.168.2.0 0.0.0.255 area 0
!
end
```

Troubleshooting Tips

If OSPF neighbors are not forming, verify:

1. IP addresses and subnet masks are correctly configured
2. Interfaces are up/up
3. OSPF process is running
4. Network statements match exactly on both routers
5. Area numbers match
6. Use debug ip ospf adj to watch neighbor adjacency formation